

**Latency-Resilient Cooperative Multi-Agent Reinforcement Learning for
Adaptive Traffic Signal Control**

Mohamed Ali

Ahmed M. S. Albreem

Mohamed Keba Njai

Abdulmajeed Mugahed Hussein Alqadami

Department of Computer Science and Engineering

Islamic University of Technology

August,

Declaration of Candidate

This is to certify that the work presented in this thesis is the outcome of the analysis and experiments carried out by **Mohamed Ali, Ahmed M. S. Albreem, Mohamed Keba Njai**, and **Abdumajeed Mugahed Hussein Alqadami** under the supervision of **Mr. S. M. Sabit Bananee**, Lecturer, Department of Computer Science and Engineering and co-supervision of **Mr. Faisal Hussain**, Assistant Professor, Department of Computer Science and Engineering, Islamic University of Technology, Dhaka, Bangladesh. It is also declared that neither this thesis nor any part of it has been submitted anywhere else for any degree or diploma. Information derived from the published and unpublished work of others have been acknowledged in the text and a list of references is given.

Mr. S. M. Sabit Bananee

Lecturer

Department of Computer Science and Engineering
Islamic University of Technology (IUT)

Date: August ,

Mohamed Ali

Student ID:

Date: August ,

Mr. Faisal Hussain

Assistant Professor

Department of Computer Science and Engineering
Islamic University of Technology (IUT)

Date: August ,

Ahmed M. S. Albreem

Student ID:

Date: August ,

Mohamed Keba Njai

Student ID:

Date: August ,

**Abdumajeed Mugahed
Hussein Alqadami**

Student ID:

Date: August ,

*Dedicated to our parents, whose love and support have been
a constant source of strength.*

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List of Abbreviations

AI	Artificial Intelligence
AIC	Akaike Information Criterion
AoI	Age of Information
BIC	Bayesian Information Criterion
CTDE	Centralized Training with Decentralized Execution
DEC-POMDP	Decentralized Partially Observable Markov Decision Process
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
GCN	Graph Convolutional Network
GNN	Graph Neural Network
IDM	Intelligent Driver Model
IPPO	Independent Proximal Policy Optimization
MADDPG	Multi-Agent Deep Deterministic Policy Gradient
MARL	Multi-Agent Reinforcement Learning
RL	Reinforcement Learning
SAFE	Stochastic Automaton Follow-the-leader Engine
SCATS	Sydney Coordinated Adaptive Traffic System
SCOOT	Split Cycle Offset Optimization Technique
SUMO	Simulation of Urban Mobility
TD	Temporal Difference
TRAFI	Traffic Flow Simulator
V2X	Vehicle-to-Everything Communication

Acknowledgement

Praise be to ALLAH, first and last, whose grace was the primary reason for success, and whose goodness and help made the completion of this research possible. To Him belongs all thanks for the blessings He granted and the ease He provided.

We extend our sincere gratitude to our supervisor, Mr. S. M. Sabit Bananee, for his endless patience, insightful critiques, and for always reminding us to “think What, and Why, How?” His expertise and encouragement were instrumental in the completion of this thesis.

We also thank our co-supervisor, Mr. Faisal Hussain, for his invaluable support and guidance throughout this research. His unwavering belief in our work sustained us, even when the challenges seemed insurmountable.

Finally, to our parents, whose love and support have been a constant source of strength: thank you for always being there and for your endless encouragement throughout our academic journey.

To all of you, we extend our heartfelt appreciation.

Abstract

Urban traffic congestion remains one of the most significant challenges faced by modern cities, causing economic losses, environmental degradation, and a reduced quality of life. Traditional traffic signal control systems rely on fixed or rule-based timing strategies that cannot adapt effectively to dynamic traffic patterns. Recent advances in Artificial Intelligence, particularly Reinforcement Learning (RL), have introduced promising approaches for adaptive traffic signal control. Among these, Multi-Agent Reinforcement Learning (MARL) allows multiple intersections to operate as intelligent agents that coordinate their decisions to optimise overall traffic flow.

Despite these advances, most existing MARL-based traffic control systems assume ideal communication conditions between agents. In real-world smart city deployments, however, communication networks introduce delays, packet loss, and asynchronous updates that can severely affect inter-intersection coordination. This gap between theoretical assumptions and practical deployment conditions remains a critical barrier to the real-world adoption of MARL-based traffic control systems.

This research proposes a latency-resilient cooperative MARL framework for adaptive traffic signal control. The proposed system systematically investigates the impact of communication latency and packet loss on cooperative traffic signal agents and explores mechanisms that allow the system to maintain stable performance under such network imperfections. The framework incorporates latency-resilient mechanisms that enable agents to use the most recent available information, estimate missing data from historical observations, and reduce their reliance on synchronous communication.

The analysis examines system behaviour under four distinct configurations: ideal communication (baseline), moderate imperfections, severe imperfections, and extreme imperfections. Performance is evaluated using established metrics, including average travel time, queue length, waiting time, and throughput. A comprehensive implementation and evaluation framework is presented. The findings provide insights into the robustness of MARL-based traffic control under realistic communication environments and contribute toward more deployable intelligent transportation systems.

Introduction

Background and Context

Urban traffic congestion is a complex, multidimensional challenge exacerbated by rapid urbanisation and the global increase in private vehicle ownership. Its economic consequences are considerable: empirical studies estimate annual losses in the billions of dollars due to diminished productivity, elevated fuel consumption, and environmental deterioration. Central to this issue is the traffic signal control system, which serves as the primary mechanism for regulating vehicular movement across urban road networks.

Conventional traffic signal control systems can be broadly classified into three generations. First-generation systems are fixed-time or pre-timed, with signal timing plans established offline based on historical traffic data. Although simple and reliable, they lack the operational flexibility to accommodate real-time traffic fluctuations. Second-generation systems are actuated, employing detectors to identify vehicle presence and adjust signal timings within predetermined constraints. While more responsive than fixed-time systems, they still rely on rule-based logic that cannot achieve network-wide optimisation. Third-generation systems are adaptive control systems, such as the Sydney Coordinated Adaptive Traffic System (SCATS) and the Split Cycle Offset Optimisation Technique (SCOOT), which dynamically modulate signal parameters in response to real-time traffic measurements. Although these represent substantial technological advances, they depend on mathematical optimisation models that may not fully capture the complex, non-linear dynamics inherent in urban traffic networks.

The emergence of Artificial Intelligence (AI) has opened new paradigms for traffic signal control. Reinforcement Learning (RL), in particular, has attracted considerable attention because it can learn optimal control policies through direct interaction with the environment, without requiring explicit mathematical models. In the RL framework, a traffic signal agent observes the state of its environment, selects an ac-

tion (signal phase), and receives a reward signal that reflects the quality of its decision. Through iterative trial-and-error, the agent learns to maximise cumulative rewards, thereby discovering effective control strategies.

When multiple intersections interact within a traffic network, the problem naturally extends to Multi-Agent Reinforcement Learning (MARL), where each intersection is modelled as an autonomous agent. These agents must learn not only to optimise their local traffic conditions but also to cooperate with neighbouring intersections to improve network-wide traffic efficiency. The MARL formulation captures the decentralised nature of real-world traffic networks, where each intersection operates with limited local observations while contributing to global system performance.

Motivation and Scope

The motivation for this research stems from a critical observation: while MARL-based approaches have shown promising results in simulated environments, their practical deployment is hindered by unrealistic assumptions about communication conditions. Most existing studies assume that agents can exchange information instantaneously and reliably—an assumption that does not hold in real-world smart city environments.

In practice, communication between traffic signal controllers typically relies on wireless networks or infrastructure-based systems. These networks are subject to various imperfections, including:

- **Communication Latency:** Delays in message delivery due to network congestion, propagation delays, or processing overhead. In vehicular networks, latency can range from tens to hundreds of milliseconds, depending on the technology and conditions.
- **Packet Loss:** The probability that transmitted messages fail to reach their destination. Packet loss rates in wireless networks vary significantly based on distance, interference, and environmental factors.
- **Asynchronous Updates:** In real-world systems, agents may operate on different timing schedules, leading to decisions based on information of varying ages and relevance.

These imperfections introduce fundamental challenges for cooperative MARL systems. When agents rely on shared information for coordination, delayed or missing data can lead to suboptimal decisions, reduced cooperation, and potentially unstable

system behaviour. Understanding the extent of these effects and developing mechanisms to mitigate them is essential for bridging the gap between theoretical MARL research and practical deployment.

This research focuses on investigating how communication imperfections influence cooperative MARL traffic signal control systems and exploring mechanisms to improve system robustness under realistic network conditions. The scope encompasses the design of a latency-resilient framework that can maintain effective coordination despite communication challenges, without requiring fundamental changes to the underlying MARL learning algorithms.

Problem Statement

Although Multi-Agent Reinforcement Learning has shown promise for adaptive traffic signal control, most existing approaches assume ideal communication environments where agents exchange information without delay or loss. This assumption creates a significant gap between simulation-based research and real-world deployment requirements.

The central problem addressed in this research can be formulated as follows: *How can cooperative MARL-based traffic signal control systems maintain effective performance when deployed in network environments characterised by communication latency, packet loss, and asynchronous updates?*

This problem encompasses several interrelated sub-problems:

1. **Impact Characterisation:** How do different levels of communication latency and packet loss affect the learning behaviour and coordination effectiveness of MARL agents?
2. **Degradation Analysis:** What is the relationship between communication imperfection severity and system performance degradation? Are there threshold levels beyond which cooperative strategies become counterproductive?
3. **Resilience Mechanisms:** What mechanisms can be incorporated into MARL frameworks to enable agents to maintain coordination despite delayed or missing information?
4. **Trade-off Analysis:** What are the trade-offs between communication resilience and other system properties, such as learning speed, convergence stability, and computational overhead?

Research Objectives

Based on the problem statement, this research pursues the following objectives:

- O1:** To systematically investigate the impact of communication latency and packet loss on cooperative MARL-based traffic signal control systems.
- O2:** To evaluate how network imperfections affect coordination mechanisms between traffic signal agents, particularly in terms of information freshness and decision synchrony.
- O3:** To propose mechanisms that improve the resilience of MARL agents under communication delays, including strategies for handling outdated information and reducing synchronous communication dependency.
- O4:** To develop a comparative framework that enables structured evaluation of system performance under ideal, degraded, and resilience-enhanced communication scenarios.
- O5:** To analyse system performance using established traffic efficiency metrics within a grounded simulation framework.

Research Challenges

The intersection of MARL and communication-constrained environments presents several fundamental challenges.

Non-Stationarity in Multi-Agent Learning

In single-agent RL, the environment is assumed to be stationary: transition dynamics and reward functions remain constant during learning. In MARL, however, multiple agents learn simultaneously, and the actions of one agent directly influence the state transitions experienced by others. This creates a non-stationary learning environment where the optimal policy for any single agent depends on the policies of all other agents. Communication delays compound this problem by forcing agents to adapt based on outdated representations of others' behaviours.

Credit Assignment in Cooperative Settings

A fundamental challenge in cooperative MARL is the credit assignment problem: how to attribute the global reward to individual agent contributions. In traffic signal con-

trol, overall traffic efficiency depends on the coordinated actions of multiple intersections, making it difficult to assess which agent's actions were most beneficial or detrimental. Communication imperfections exacerbate this issue by introducing uncertainty about the actual state information available to each agent at decision time.

Exploration–Exploitation in Cooperative Contexts

Each agent must balance exploration (trying new actions to discover better strategies) with exploitation (using known good strategies). In cooperative settings, one agent's exploration may disrupt the learning of others, creating conflicts between individual and collective exploration strategies. Communication delays further complicate this balance, as agents may be unable to coordinate their exploration effectively.

Communication Imperfection Modelling

Accurately modelling communication imperfections requires careful consideration of:

- **Stochastic latency** — not a fixed delay, but varying with network conditions.
- **Correlated packet loss** — losses may be correlated over time or space, not independent.
- **Interplay between latency and loss** — complex dynamics not captured by simple additive models.

Developing abstractions that capture the essential characteristics of real-world communication while remaining tractable for analysis is a significant challenge.

Scalability Considerations

As the number of intersections increases, communication overhead grows, and the effects of latency and packet loss may become more pronounced. Ensuring that the proposed resilience mechanisms scale effectively to larger networks is essential for practical applicability.

Contributions

This research makes six contributions to the field of intelligent traffic signal control using Multi-Agent Reinforcement Learning, with particular emphasis on robustness

under communication constraints.

- C1: Theoretical Framework for Communication Imperfection Modelling in MARL.** A formal framework incorporates communication latency and packet loss into the MARL formulation for traffic signal control, providing formal models for delayed and unreliable message exchange between agents. This provides a foundation for systematic analysis of communication effects on cooperative learning.
- C2: Latency-Resilient Cooperative MARL Design Principles.** Design principles are established for latency-resilient cooperative MARL systems, including information freshness management, graceful degradation under communication stress, and reduced dependency on synchronous communication. The principles apply to various MARL algorithms and traffic network configurations.
- C3: Structured Comparative Analysis Framework.** A structured experimental framework evaluates system performance under three conditions: ideal communication, communication with latency and packet loss, and latency-resilient MARL.
- C4: Cooperative Reward Design for Network-Wide Optimisation.** A cooperative reward mechanism balances local intersection performance with global traffic efficiency, addressing selfish behaviour in decentralised MARL systems while maintaining local responsiveness.
- C5: Analysis of Communication–Performance Relationships.** The relationships between communication parameters (latency magnitude, packet loss rate) and system performance metrics are analysed, identifying critical thresholds and degradation patterns that guide practical system design.
- C6: Bridging Theory and Practice in Smart City MARL.** More broadly, this research contributes to robust Multi-Agent Reinforcement Learning by demonstrating that communication constraints can be addressed without modifying the underlying learning algorithm.

Thesis Organisation

The remainder of this thesis is structured as follows:

Literature Review presents a comprehensive literature review and derives the four research gaps (G1–G4) that motivate the thesis.

Methodology presents the proposed methodology: the formal system model (closing G1), the degradation model (informing G2), and the latency-resilience mechanisms M1–M5 (closing G3).

System Implementation and Experimental Framework presents the system implementation and experimental framework, including the layered software architecture, communication impairment injection, baselines, configurations, metrics, and training protocol that together close G4.

Results and Analysis presents the results and key findings F1–F5, quantifying the impact of imperfections (G2) and the effectiveness of the resilience mechanisms (G3).

Discussion provides a critical discussion, including explicit closure of each research gap and the study limitations.

Conclusion concludes, summarises contributions, and organises future research into a roadmap.

Research Thread: From Gaps to Findings

To keep the argument of this thesis transparent, every key element is cross-referenced throughout: the research gaps identified in the literature review are labelled G1–G4; the objectives O1–O5; the contributions C1–C6; the design principles P1–P3; the resilience mechanisms M1–M5; and the key findings F1–F5. Table 1 merges these key points into a single research thread, showing how each gap is addressed and where its closure is validated.

Table 1: The Research Thread: How Each Identified Gap Is Addressed and Validated

Research Gap	Objective	Contribution Mechanism / Element		Validated by
G1: No formal model of communication imperfections in traffic MARL	O1	C1	P1; delay/loss-augmented DEC-POMDP with AoI observations (Eqs. 1–4)	Ch. 3; F2
G2: Impact of latency/loss on cooperative traffic MARL unquantified	O1, O2	C5	Logistic degradation model (Eq. 3); configurations C1–C4	Ch. 5; F2, F3
G3: No resilience mechanisms for stale/missing information	O3	C2, C4	P2, P3; mechanisms M1–M5; cooperative reward (Eq. 6)	Ch. 5; F4, F5
G4: No structured evaluation methodology under realistic communication	O4, O5	C3	Implemented system; five baselines; four configurations; metrics suite; paired protocol	Ch. 4; F1, F5

The remainder of the thesis follows this thread: Chapters 2–4 develop the elements of the middle column, Chapter 5 produces the findings of the last column, and Chapter 6 explicitly assesses the closure of G1–G4.

Literature Review

Introduction to the Literature Landscape

The application of Reinforcement Learning to traffic signal control has evolved through several distinct phases, each building on the foundations established by previous work. This chapter provides a comprehensive review of the relevant literature, organised thematically to highlight the progression of ideas and identify gaps that motivate the current research. The review encompasses single-agent RL approaches, cooperative MARL frameworks, communication-aware multi-agent systems, and resilience mechanisms in distributed systems. The goal is to clearly map the field of communication-resilient MARL for traffic control, distinguishing what has been achieved from what remains unexplored.

Foundations: Single-Agent Reinforcement Learning for Traffic Signal Control

Early applications of RL to traffic signal control focused on single-agent formulations, where each intersection is treated as an independent learning entity. This approach, while limited in capturing network-wide interactions, established important foundations for later multi-agent extensions.

Early Q-Learning Approaches

The pioneering work of Thorpe and Anderson [11] applied Q-learning to traffic signal control, demonstrating that RL agents could learn effective signal timing policies through interaction with simulated traffic environments. Their approach used a simple state representation based on queue lengths and vehicle counts, with actions corresponding to different signal phase selections. The reward function penalised queue lengths and rewarded vehicle throughput, providing a clear optimisation objective.

Subsequent work by Abdulhai et al. [1] extended Q-learning to adaptive traffic signal control, introducing more sophisticated state representations that included information about approaching vehicles. Their results showed that RL-based controllers could outperform fixed-time systems, particularly under variable traffic conditions.

Deep Reinforcement Learning Approaches

The advent of Deep Reinforcement Learning (DRL) enabled more powerful approaches that could handle high-dimensional state spaces. Wei et al. [12] applied Deep Q-Networks (DQN) to traffic signal control in the IntelliLight system, using raw traffic data as input to neural networks that approximated the Q-function. This approach eliminated manual feature engineering and allowed agents to learn from richer state representations. A comprehensive survey by Wei et al. [13] later summarised the evolution of RL methods in this domain.

Van der Pol and Oliehoek [8] introduced Deep Q-learning with a pressure-based state representation, where the state was defined as the difference between incoming and outgoing traffic pressure at each intersection. This representation provided a more physically meaningful state encoding and improved learning efficiency.

Li et al. [4] proposed G2ANet, a graph attention network approach that enabled agents to aggregate information from neighbours with learned attention weights, demonstrating the importance of structured communication for scalable multi-agent learning.

Limitations of Single-Agent Approaches

While single-agent RL approaches demonstrated the potential of learning-based traffic control, they suffer from fundamental limitations. First, by treating each intersection independently, they fail to capture the interdependencies between neighbouring intersections. Traffic flows propagate through networks, and decisions at one intersection directly affect conditions at adjacent ones. Second, single-agent approaches optimise only for local performance, which may lead to globally suboptimal outcomes. Third, their scalability is limited, as adding more intersections requires training additional independent agents without any coordination mechanism.

These limitations motivated the shift toward Multi-Agent Reinforcement Learning, where intersections are modelled as cooperating agents that can share information and coordinate their decisions.

Cooperative Multi-Agent Reinforcement Learning for Traffic Control

The transition from single-agent to multi-agent formulations introduced new challenges and opportunities. This section reviews the evolution of cooperative MARL approaches for traffic signal control, emphasising the communication models employed.

Independent Learning Approaches

The simplest MARL extension treats each agent as an independent learner, using single-agent RL algorithms without explicit coordination mechanisms. While easy to implement, independent learning suffers from non-stationarity, as each agent's learning environment changes as other agents update their policies.

Wiering [14] extended Q-learning to multi-agent traffic control using a simple independent learning approach. Despite the lack of explicit coordination, the agents achieved reasonable performance due to implicit coordination emerging from shared environmental dynamics. However, the approach showed instability and high variance in performance across training runs. More recent independent learning frameworks, such as IPPO (Independent PPO), have been applied to traffic networks, confirming that while they can be competitive, they often underperform coordinated methods when network interdependencies are strong.

Centralized Training with Decentralized Execution (CTDE)

The CTDE paradigm has emerged as the dominant approach for cooperative MARL. In CTDE, agents have access to global information during training (enabling centralised critics or value functions) but must act based only on local observations during execution. This approach balances the benefits of centralised coordination during learning with the practical requirements of decentralised deployment.

Value Decomposition Methods

Value decomposition methods address the challenge of learning a joint value function that can be factored into individual agent value functions. QMIX by Rashid et al. [9] introduced a monotonic mixing network that combines individual agent Q-values into a global Q-value while maintaining the individual-global-max property. This enables

centralised training with guaranteed consistency between individual and global value estimates.

In the traffic control domain, value decomposition has been applied to enable intersections to learn cooperative policies. The joint Q-value represents the expected cumulative reward for the entire network, while individual Q-values guide decentralised decision-making. Variants such as QTRAN and Weighted QMIX have been proposed to relax monotonicity constraints, allowing more complex coordination patterns.

Actor-Critic Methods

Multi-Agent Deep Deterministic Policy Gradient (MADDPG) by Lowe et al. [6] extended the actor-critic framework to multi-agent settings. Each agent has its own actor (policy network) but shares a centralised critic that evaluates actions based on the joint state and action of all agents. During execution, each actor selects actions based only on local observations.

MADDPG and its variants have been applied to traffic signal control, enabling agents to learn coordinated policies through centralised criticism while maintaining decentralised execution. The approach handles continuous action spaces, which is relevant for traffic control scenarios where signal phase durations can be treated as continuous variables. Extensions such as MATD3 and MASAC have improved stability and sample efficiency in traffic applications.

Communication-Based Cooperation

Explicit communication between agents provides an alternative mechanism for coordination. In communication-based approaches, agents can transmit messages containing selected information to neighbours, enabling more direct coordination than implicit cooperation through shared rewards.

Learned Communication

CommNet by Sukhbaatar et al. [10] introduced a framework where agents learn to communicate through continuous vectors. Each agent receives messages from all other agents and incorporates this information into its decision-making process. The communication protocol itself is learned end-to-end alongside the control policy.

Das et al. [2] introduced targeted communication where agents selectively communicate with relevant neighbours, reducing communication overhead while maintaining coordination benefits.

Graph-Based Communication

Graph Neural Networks (GNNs) provide a natural framework for modelling communication in traffic networks, where intersections can be represented as nodes and communication links as edges. G2ANet by Li et al. [4] applied graph attention networks to traffic signal control, enabling agents to aggregate information from neighbours with learned attention weights. Subsequent works have used Graph Convolutional Networks (GCNs) to propagate traffic state information across the network, showing improved scalability.

Analysis of Existing Cooperative Approaches

A critical observation from the literature is that most cooperative MARL approaches for traffic control assume ideal communication conditions. When communication is modelled explicitly, it is typically assumed to be instantaneous and reliable. For example, in CTDE approaches, the centralised critic is assumed to have access to complete and current information from all agents. In communication-based approaches, messages are assumed to be delivered without delay or loss.

This assumption may be reasonable in simulated environments where all agents operate on the same computational platform and share memory. However, it does not reflect the reality of distributed deployments where agents are physically separated and communicate through networks subject to latency and packet loss. Consequently, the field of MARL for traffic control still lacks a systematic treatment of communication imperfections, leaving a significant gap between laboratory performance and real-world viability.

Communication-Aware Multi-Agent Systems

A growing body of research recognises the importance of communication constraints in multi-agent systems, though this work has largely developed outside the traffic control domain. This section maps the key concepts that are transferable to our problem.

Delayed Reinforcement Learning

Research on delayed rewards in RL provides relevant insights. In settings where rewards are received with delay, agents must learn to associate actions with outcomes that may be temporally distant. Similarly, in communication-delayed settings, agents

must make decisions based on state information that may not reflect the current situation.

Joulani et al. [3] provided theoretical analysis of delayed reward RL, establishing bounds on the performance degradation caused by reward delays. Their work suggests that moderate delays can be tolerated if agents maintain appropriate buffers and update strategies. In multi-agent settings, communication delays further compound this problem because each agent's observation of others is delayed, effectively creating a delayed, partially observable Markov game.

Asynchronous Multi-Agent Learning

Asynchronous multi-agent learning addresses settings where agents operate on different timing schedules. A3C (Asynchronous Advantage Actor-Critic) by Mnih et al. [7] demonstrated that asynchronous learning can improve training efficiency in single-agent settings by enabling multiple parallel learners.

In multi-agent settings, asynchrony introduces additional challenges. Agents may be at different stages of their learning process, and their policies may be based on different amounts of experience. Communication delays exacerbate this asynchrony by introducing uncertainty about the currency of received information. Recent work on asynchronous MARL, such as asynchronous QMIX, has shown that naively ignoring timing differences can destabilise training, motivating the need for explicit temporal alignment mechanisms.

Robust Communication in Distributed Systems

Research on robust communication in distributed systems provides principles that can inform the design of resilient MARL frameworks. Techniques such as message buffering, timestamping, and stale-data detection have been developed to handle communication imperfections in traditional distributed systems.

Consensus algorithms in distributed computing, such as Paxos and Raft, provide mechanisms for achieving agreement among distributed agents despite communication failures. While not directly applicable to MARL (which does not require exact consensus), these algorithms demonstrate that robust coordination is possible under communication constraints. Additionally, information freshness metrics such as Age of Information (AoI) have been studied in networked control systems; these concepts are directly relevant to weighting received messages based on timeliness.

Communication-Efficient and Robust MARL

Recent efforts in the broader MARL community have begun to address communication limitations. For example, work on “When to Communicate” (e.g., IC3Net, TarMAC) decides when and what to communicate to reduce bandwidth. However, these approaches still assume that once a message is sent, it arrives instantly and reliably. Studies on robust MARL against adversarial attacks or environment noise have emerged, but the specific case of stochastic communication latency and packet loss in cooperative traffic control remains largely unexplored.

Gap Analysis: Communication Imperfections in Traffic MARL

Despite the extensive literature on MARL for traffic control and communication-aware multi-agent systems, a critical gap exists: the systematic study of how communication latency and packet loss affect cooperative MARL-based traffic signal control.

Assumptions in Existing Work

A systematic review of the traffic control MARL literature reveals four consistent assumptions about communication:

- A1: Instantaneous Information Sharing:** Most CTDE approaches assume that centralised components have immediate access to all agent information.
- A2: Reliable Message Delivery:** Communication-based approaches typically assume that all transmitted messages are received successfully.
- A3: Synchronised Decision-Making:** Agents are assumed to make decisions at synchronised time steps, with no temporal offset between agents.
- A4: Homogeneous Communication Quality:** When communication quality is considered, it is typically assumed to be uniform across all agent pairs.

Consequences of Unrealistic Assumptions

These assumptions create a gap between theoretical performance (observed in simulations) and expected practical performance (in real-world deployments). Agents trained under ideal communication conditions may exhibit significant performance degradation when deployed in environments with realistic communication constraints.

Furthermore, the lack of consideration for communication imperfections may lead to over-optimised solutions that are fragile to small perturbations. Robustness to communication variations should be a design consideration, not an afterthought.

Synthesis: The Four Research Gaps

The assumptions A1–A4, their consequences, and the unexplored transfer of communication-aware concepts to the traffic domain consolidate into four research gaps that define the scope of this thesis:

- G1: Formal modelling gap.** No formal framework incorporates communication latency and packet loss into the MARL formulation for traffic signal control; observations treat received messages as current and delivered, leaving the AoI of coordination information unmodelled.
- G2: Impact quantification gap.** The degradation caused by different severities of latency and packet loss on cooperative traffic MARL is uncharacterised; no critical thresholds are known beyond which cooperative strategies become counterproductive.
- G3: Resilience gap.** No mechanisms exist that allow agents to remain coordinated with stale or missing information without fundamentally altering the learning algorithm.
- G4: Evaluation gap.** No structured methodology—baselines, communication configurations, and metrics—exists for evaluating MARL robustness under realistic communication conditions.

Opportunities for Contribution

The identified gaps present clear opportunities, each mapped to the contributions C1–C6 through the research thread of Table 1:

1. **Impact Analysis** (addresses G2): systematically analysing how different types and severities of communication imperfections affect system performance.
2. **Resilience Design** (addresses G3): proposing design principles and mechanisms for improving resilience to communication challenges.
3. **Validation Framework** (addresses G4): establishing frameworks for evaluating system robustness under realistic communication conditions.

Summary and Research Positioning

The literature review establishes the working field of this thesis as follows:

1. MARL has demonstrated strong potential for traffic signal control, outperforming traditional methods in simulated environments.
2. Cooperative mechanisms (CTDE, communication-based, graph-based) enable effective multi-intersection coordination, but they universally assume perfect communication (A1–A4), giving rise to gaps G1–G4.
3. Research on communication-aware multi-agent systems provides foundational concepts (delayed rewards, AoI, robust consensus), yet these have not been systematically applied to cooperative traffic control MARL.
4. There is a clear need for frameworks that explicitly address communication imperfections, combining formal modelling (G1), resilience mechanisms (G3), and a structured evaluation methodology (G4).

This research addresses the identified gaps by proposing a latency-resilient cooperative MARL framework that explicitly incorporates communication imperfections into the system design and evaluation. The following chapters present the theoretical foundations, methodology, and analysis of this approach, following the research thread summarised in Table 1.

Methodology

Overview of the Proposed Framework

This chapter presents the theoretical foundation of the proposed latency-resilient co-operative Multi-Agent Reinforcement Learning (MARL) framework for adaptive traffic signal control. In contrast to conventional designs that treat the communication network as an idealised, lossless, and instantaneous medium, the proposed framework treats communication imperfections as first-class elements of the system model. Robustness to latency and packet loss is therefore achieved *by design*, rather than through post-hoc compensation. The formal model developed in this chapter closes gap G1; the mechanisms of Section close gap G3.

The methodology rests on three design principles, referenced throughout the thesis as P1–P3:

- P1 Explicit Imperfection Modelling.** Communication latency and packet loss are incorporated into the formal problem definition from the outset, so that the gap between theoretical assumptions and deployment conditions is closed at the modelling level.
- P2 Graceful Degradation.** The system is engineered to lose performance smoothly as communication quality deteriorates, preserving acceptable control behaviour instead of failing catastrophically beyond a critical imperfection threshold.
- P3 Freshness-Aware Decision Making.** Agents explicitly track the Age of Information (AoI) of every received message, discount stale data, compensate for missing data through principled estimation, and thereby reduce their dependency on synchronous communication.

Table 2 summarises how each principle responds to a specific challenge identified in Chapters 1 and 2 and is operationalised by a concrete mechanism.

Table 2: Correspondence between Design Principles, Challenges, and Realising Mechanisms

Principle	Challenge Addressed	Realising Mechanism
P1	Ideal-channel assumptions (G1)	Delay/loss-augmented DEC-POMDP (M1)
P2	Coordination collapse under stress (G2)	Cooperative reward; dropout training (M4, M5)
P3	Decisions based on stale data (G3)	AoI weighting; state prediction (M2, M3)

Formal Problem Formulation

Traffic Network as a Decentralized POMDP

The control problem under imperfect communication is formalised as a Decentralised Partially Observable Markov Decision Process (DEC-POMDP):

$$\mathcal{G} = \langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, \{\mathcal{O}_i\}, T, R, \{\Omega_i\}, \gamma \rangle, \quad (1)$$

where $\mathcal{N} = \{1, \dots, n\}$ is the set of n signalised intersections (agents); $\mathcal{S}$ is the set of global network states; $\mathcal{A}_i$ and $\mathcal{O}_i$ are the action and observation sets of agent i ; $T : \mathcal{S} \times \mathcal{A}_1 \times \dots \times \mathcal{A}_n \rightarrow \Delta(\mathcal{S})$ is the state-transition function; $R : \mathcal{S} \times \mathcal{A}_1 \times \dots \times \mathcal{A}_n \rightarrow \mathbb{R}$ is the shared team reward; $\Omega_i : \mathcal{S} \times \mathcal{A}_i \rightarrow \Delta(\mathcal{O}_i)$ is the observation function of agent i ; and $\gamma \in [0, 1)$ is the discount factor.

Each agent i seeks a policy $\pi_i : \mathcal{O}_i \times \mathcal{M}_i \rightarrow \Delta(\mathcal{A}_i)$ that maps its local observation and the set of received messages $\mathcal{M}_i$ to a distribution over actions, with the collective objective of maximising the expected discounted team reward $J = \mathbb{E} [\sum_{t=0}^{\infty} \gamma^t r^t]$.

[Placeholder: 3×3 grid network diagram]

Figure 1: A 3×3 grid traffic network topology consisting of nine signalised intersections. The grid topology is a standard benchmark in traffic control literature.

The traffic network topology used throughout this work is shown in Figure 1.

Communication-Augmented Observation Space

To render communication imperfections *observable*—and thereby close gap G1—the observation of agent i at time t is extended with message metadata:

$$o_i^t = \langle s_i^{\text{local},t}, \{m_{j \rightarrow i}^{t-\tau_{ji}}, \tau_{ji}\}_{j \in \mathcal{N}_i^{\text{rec}}} \rangle, \quad (2)$$

where $s_i^{\text{local},t}$ is the local observation at time t ; $m_{j \rightarrow i}^{t-\tau_{ji}}$ is the message sent by neighbour j at time $t - \tau_{ji}$; τ_{ji} is the incurred delay (the AoI of the message); and $\mathcal{N}_i^{\text{rec}} \subseteq \mathcal{N}_i$ is the subset of neighbours whose latest messages were successfully delivered. Because τ_{ji} is part of the observation itself, the policy can condition its decisions on information freshness—an ability that standard MARL formulations, which implicitly assume A1 and A2, do not provide.

Simulation Environment and Traffic Dynamics

To accurately evaluate the proposed latency-resilient MARL framework, a robust simulation environment with realistic microscopic traffic dynamics is essential. This section details the selection of the underlying car-following model and the mathematical formulation of traffic state evolution.

Car-Following Model Selection and Empirical Validation

The behaviour of individual vehicles significantly impacts the overall traffic state, particularly when agents make suboptimal decisions due to communication delays. The TRAFI (Traffic Flow) Simulator is used to model the grid network. To select the most appropriate microscopic traffic model, three standard car-following models were benchmarked under a baseline free-flow scenario (44 vehicles, road length 190m, step size 0.35s).

The comparative results of the baseline simulation are presented in Table 3.

Table 3: Comparison of Car-Following Models in TRAFI Simulator

Model	Avg Travel Time (s)	Avg Speed (m/s)	System Efficiency (%)	Avg Delay (s)
GIPPS	114.77	18.96	94.3%	6.54
IDM	113.19	19.20	95.6%	4.96
SAFE	112.39	19.33	96.3%	4.16

The SAFE (Stochastic Automaton Follow-the-leader Engine) model demonstrates superior performance, achieving the lowest average delay (4.16 s) and the highest system efficiency (96.3%). While the baseline scenario exhibits zero waiting time (indicating free-flow conditions), the higher fidelity of the SAFE model provides a more realistic representation of stop-and-go dynamics. The SAFE model is therefore selected as the

default car-following mechanism for the simulation environment to ensure that the MARL agents are trained and evaluated under realistic vehicular kinematics.

Empirical Traffic Flow and Degradation Modelling

While the baseline simulation represents free-flow conditions, the introduction of communication latency and packet loss inevitably leads to localised congestion and queue formation. To mathematically capture the transition from free-flow to congested states—and to model how network imperfections degrade intersection performance—a logistic growth model is employed:

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}}, \quad (3)$$

where x represents the independent variable (e.g., communication latency in milliseconds or traffic density); L is the carrying capacity representing the maximum saturation limit (e.g., maximum queue length or maximum tolerable delay); k is the growth rate parameter determining the steepness of the degradation curve; and x_0 is the inflection point at which the system begins to saturate rapidly.

Based on empirical data fitting derived from preliminary congestion simulations, the derived parameters are $L = 302.23$, $k = 0.0143$, and $x_0 = 101.54$. To validate the selection of the logistic model, it was compared against polynomial and exponential baselines using the Akaike Information Criterion (AIC) and the coefficient of determination (R^2). As shown in Table 4, the logistic model achieves the highest R^2 (0.9861) and the lowest AIC (4043.81), confirming it as the most parsimonious and accurate representation of traffic state degradation.

Table 4: Comparison of Curve Fitting Models for Traffic Degradation Behaviour

Model	R^2	AIC	BIC
Logistic	0.9861	4043.81	4058.47
Polynomial (deg 2)	0.8892	6080.37	6095.03
Polynomial (deg 1)	0.5376	7478.30	7488.08
Exponential	0.4736	7605.22	7614.99

This mathematical model directly addresses gap G2: it provides the quantitative law relating imperfection severity to performance. The inflection point $x_0 = 101.54$ represents a critical threshold (in milliseconds of latency) beyond which coordination

rapidly fails and queues grow toward the physical capacity $L = 302.23$. The proposed latency-resilient MARL architecture specifically targets operation around this inflection point, ensuring that agents use freshness-aware mechanisms to prevent the system from crossing into the saturated degradation regime.

Communication Model

Latency Model

The communication delay between agents j and i is modelled as a random variable $\tau_{ji} \sim \mathcal{D}_\tau$, where $\mathcal{D}_\tau$ may be a fixed delay ($\tau_{ji} = \tau_0$), a uniform distribution, or an exponential distribution $\text{Exp}(\lambda_\tau)$. The exponential variant models memoryless network congestion and is used for the severe and extreme configurations of Section .

Packet Loss Model

Message delivery failure is modelled as a Bernoulli trial, $L_{ji}^t \sim \text{Bernoulli}(p_{\text{loss}})$, where p_{loss} is the probability that the message from agent j to agent i is lost at step t .

Combined Model

The effective message set available to agent i at time t is

$$\mathcal{M}_i^t = \{m_{j \rightarrow i}^{t-\tau_{ji}} : j \in \mathcal{N}_i, L_{ji}^t = 0\}. \quad (4)$$

Equations 2 and 4 jointly define the delay- and loss-augmented observation process through which design principle P1 is realised and assumptions A1–A2 are relaxed within the model.

[Placeholder: framework architecture diagram]

Figure 2: Architectural overview of the latency-resilient CTDE framework. During execution, agents communicate through imperfect channels. During training, the centralised critic uses all local and delayed observations to evaluate actions.

The overall system architecture is illustrated in Figure 2.

Agent Architecture Design

State Representation

Each agent's state representation captures both local traffic conditions and, when available, information from neighbouring agents. The state vector for agent i at time t consists of the following.

Local Traffic Features:

- Queue length for each incoming lane: $q_{i,\ell}^t$ for lane ℓ
- Average waiting time for each incoming lane: $w_{i,\ell}^t$
- Current signal phase: ϕ_i^t
- Time elapsed in current phase: δ_i^t

Neighbour Information (when available):

- Received queue lengths from neighbours: $\hat{q}_j^{t-\tau_{ji}}$
- Received phase information from neighbours: $\hat{\phi}_j^{t-\tau_{ji}}$
- Age of received information: τ_{ji}

Action Space

The action space for each agent is defined as a discrete set of traffic signal control decisions:

$$\mathcal{A}_i = \{ \text{keep current phase, switch to phase 1, ... , switch to phase } |\Phi_i| - 1 \}, \quad (5)$$

where Φ_i denotes the set of signal phases at intersection i .

Neural Network Architecture

Each agent is implemented as a parameterised policy π_{θ_i} composed of four functional modules:

1. **Feature-Extraction Module.** Fully connected layers encode the raw local features (queue lengths, waiting times, current phase, elapsed phase time) into a latent representation h_i^t .

2. **Freshness-Aware Aggregation Module.** Neighbour messages are combined through attention weights that are discounted by message age, as formalised in mechanism M2 (Section).
3. **Communication-Dropout Module.** A stochastic mask simulates packet loss during training, regularising the policy against missing neighbour information (M5).
4. **Policy Head.** A categorical output layer produces a distribution over the discrete action set $\mathcal{A}_i$.

Cooperative Reward Design

The reward balances local responsiveness with network-wide coordination:

$$r_i^t = \beta r_{\text{local},i}^t + (1 - \beta) r_{\text{global}}^t, \quad (6)$$

with

$$r_{\text{local},i}^t = - \sum_{\ell \in \mathcal{L}_i} q_{i,\ell}^t, \quad r_{\text{global}}^t = - \frac{1}{n} \sum_{j \in \mathcal{N}} \bar{q}_j^t, \quad (7)$$

where $\mathcal{L}_i$ is the set of incoming lanes at intersection i ; $\bar{q}_j^t$ is the mean queue length at intersection j ; and $\beta \in (0, 1)$ is the local-global trade-off coefficient. The global term implements the network-wide objective, while the local term preserves responsiveness when communication degrades (principle P2). Setting β strictly between 0 and 1 discourages selfish phase selection while retaining sensitivity to local demand.

Latency-Resilience Mechanisms

This section operationalises principles P1–P3 through five mechanisms, M1–M5, which collectively close gap G3. Notably, the degradation model of Section 3.3.2 identifies the inflection point x_0 beyond which coordination collapses; the mechanisms below are designed to keep the effective information state of the system on the resilient side of this threshold. Each mechanism is presented in a uniform structure: *Rationale*, *Operation*, and *Effect*.

M1: Freshness-Aware Observation Encoding

Rationale. Standard MARL observations conflate current and stale neighbour information, leaving the policy unable to distinguish between them.

Operation. Every broadcast message carries a transmission timestamp. Upon arrival, the AoI τ_{ji} is computed and appended to the observation vector, yielding Eq. 2.

Effect. The policy can learn age-dependent trust: recent messages inform fine-grained coordination, whereas old messages serve only as weak evidence.

M2: AoI-Weighted Attention Aggregation

Rationale. Naively averaging neighbour information lets a single badly delayed message contaminate the aggregated state.

Operation. Each neighbour contribution is discounted by a monotone decay kernel before attention aggregation:

$$\tilde{m}_{j \rightarrow i}^t = \kappa(\tau_{ji}) m_{j \rightarrow i}^{t - \tau_{ji}}, \quad \kappa(\tau) = e^{-\lambda \tau}, \quad (8)$$

where the decay rate λ is learned jointly with the policy, allowing the system to discover its effective information horizon end-to-end.

Effect. Stale messages are attenuated smoothly rather than discarded abruptly, yielding graceful degradation (P2) instead of discontinuous behaviour at the freshness boundary.

M3: Predictive Stale-State Compensation

Rationale. When messages are delayed or lost, the agent's view of its neighbours diverges from reality; discarding the history wastes useful evidence.

Operation. A learned forward model g_ψ extrapolates the neighbour state from the last received message and its age:

$$\hat{s}_j^t = g_\psi(m_{j \rightarrow i}^{t - \tau_{ji}}, \tau_{ji}). \quad (9)$$

Effect. The effective information horizon is extended beyond the last successful delivery, reducing the cost of packet loss and long delays.

M4: Asynchronous Communication Cadence

Rationale. Synchronous round-based protocols amplify latency costs, because an entire round stalls while waiting for the slowest message.

Operation. Agents broadcast on a fixed local cadence and maintain a most-recent-wins buffer per neighbour, which always supplies the newest delivered message with its timestamp; no agent ever blocks on the network.

Effect. Decision-making is decoupled from the communication schedule, directly reducing dependency on synchronous communication (P3) and bounding the impact of any single missing message.

M5: Communication-Dropout Training

Rationale. A policy trained exclusively under ideal communication is brittle when messages vanish at deployment time.

Operation. During centralised training, neighbour messages are randomly masked with probability p_{drop} , matched to the anticipated deployment loss rate.

Effect. The policy is regularised to remain functional across the full range of communication conditions—robustness by training (P2).

Learning Algorithm

The framework adopts the Centralised Training with Decentralised Execution (CTDE) paradigm, adapted to delayed and lossy information flow: the centralised critic consumes all agents' trajectories (possibly with their original delays), while each actor conditions only on its communication-augmented observation. Algorithm 1 summarises the procedure.

Algorithm 1 Latency-Resilient CTDE Training Procedure

```
1: Initialise actors  $\{\theta_i\}$ , centralised critic  $\phi$ , forward model  $g_\psi$ 
2: for each training episode do
3:   Reset environment, message buffers, and AoI registers
4:   for each decision step  $t$  do
5:     Every agent broadcasts  $m_i^t$  through the imperfect channel
6:     Apply communication dropout mask with probability  $p_{\text{drop}}$  ▷ M5
7:     Build  $o_i^t$ : encode AoI ▷ M1, aggregate with decay ▷ M2, compensate
       missing states ▷ M3
8:     Execute  $a_i^t \sim \pi_{\theta_i}(\cdot \mid o_i^t)$ ; observe  $r^t$ , next state
9:     Update per-neighbour most-recent-wins buffers ▷ M4
10:   end for
11:   Update critic  $\phi$  on collected (delayed) trajectories
12:   Update actors  $\{\theta_i\}$  and decay rate  $\lambda$  via policy gradient using Eq. 6
13:   Update  $g_\psi$  by supervised regression on observed transitions
14: end for
```

Chapter Summary

This chapter formalised traffic signal control under imperfect communication as a delay- and loss-augmented DEC-POMDP (**G1 closed**), introduced the empirical logistic degradation law with its critical threshold $x_0 \approx 102$ ms (**G2 informed**), designed a cooperative reward balancing local and global objectives, and developed five latency-resilience mechanisms M1–M5 (**G3 closed**). The next chapter presents the concrete software implementation of this framework and the experimental methodology that closes G4.

System Implementation and Experimental Framework

Overview of the Implementation

Chapter 3 developed the latency-resilient cooperative MARL framework theoretically; this chapter presents its concrete realisation as an executable system, together with the experimental framework through which it is evaluated. The implementation follows a four-layer architecture in which each layer owns a distinct responsibility and interacts with its neighbours through well-defined interfaces:

- L1 Traffic Simulation Layer.** Microscopic vehicle dynamics on the 3×3 grid network, using the TRAFI simulator with the SAFE car-following model selected in Section 3.3.1.
- L2 Communication Impairment Layer.** A channel emulator that injects configurable latency and packet loss into every inter-agent message, implementing the formal model of Section .
- L3 Agent Runtime Layer.** The policy networks and the software components realising mechanisms M1–M5: AoI registers, decay-weighted aggregation, the predictive forward model, most-recent-wins buffers, and the communication-dropout masker.
- L4 Learning Layer.** The CTDE training loop: experience replay, the centralised critic, and the supervised updates of the forward model g_ψ .

Figure 3 depicts the architecture, and Table 5 maps each layer to its responsibility and realisation.

[Placeholder: four-layer software architecture diagram]

Figure 3: Four-layer software architecture of the implemented system. Agents in the runtime layer (L3) observe the traffic simulation (L1) through the impairment channel (L2) and are trained by the learning layer (L4) under the CTDE paradigm.

Table 5: Layered Architecture: Responsibilities and Realisations

Layer	Responsibility	Realisation	Section
L1	Microscopic traffic dynamics	TRAFI simulator, SAFE model	§
L2	Latency and loss injection	Scheduled delivery queue	§
L3	Policies and mechanisms M1–M5	Neural modules + message buffers	§
L4	CTDE training	Replay buffers, centralised critic	§

Simulation Platform and Network Model

The traffic environment is implemented in the TRAFI (Traffic Flow) Simulator, configured with the SAFE car-following model as the environment default (Section 3.3.1). The network is the 3×3 grid of Figure 1 with nine signalised four-way intersections, two through-lanes per approach, 300 m road segments, a free-flow speed of 50 km/h, and four-phase signal plans with 10–60 s green bounds, 3 s yellow intervals, and a 10 s agent decision interval.

Four demand scenarios span the operating regimes of interest (Table 6). Fixed-time and actuated control serve as communication-independent anchors, since their logic involves no inter-agent messaging.

Table 6: Traffic Demand Scenarios

Scenario	Description	Base Demand	Directional Bias
S1	Uniform flow	300 veh/h/approach	none
S2	Morning peak	450 veh/h/approach	60% inbound
S3	Evening peak	450 veh/h/approach	60% outbound
S4	Oversaturated	600 veh/h/approach	none

Remark on external validation. SUMO [5] is retained as a complementary platform for cross-simulator validation: its TraCI interface exposes the same decision-step semantics, so the agent runtime (L3) and learning layer (L4) are simulator-agnostic and can be attached to either platform without modification.

Communication Impairment Injection Layer

Every inter-agent message is routed through an impairment channel that implements the formal model of Section . Messages carry a fully defined structure:

$$m_{j \rightarrow i}^t = \langle j, i, t, \sigma_j, \underbrace{(\phi_j^t, \mathbf{q}_j^t, \delta_j^t)}_{\text{payload}} \rangle, \quad (10)$$

where t is the transmission timestamp, σ_j is a monotonically increasing per-agent sequence number, and the payload contains the sender's current phase ϕ_j^t , per-lane queue vector $\mathbf{q}_j^t$, and elapsed phase time δ_j^t . The sequence number enables most-recent-wins buffering (M4) and the timestamp enables AoI computation (M1).

Algorithm 2 specifies the channel operation. Latency draws $\tau \sim \mathcal{D}_\tau$ and loss draws are produced by dedicated, seeded random number generators so that any experiment is exactly reproducible.

Algorithm 2 Communication Impairment Injection Channel

Require: message $m = \langle j, i, t, \sigma_j, \text{payload} \rangle$; parameters $\mathcal{D}_\tau, p_{\text{loss}}$; receiver buffer B_i

- ```

1: Draw loss indicator $L \sim \text{Bernoulli}(p_{\text{loss}})$
2: if $L = 1$ then
3: drop m and record a loss event ▷ packet loss
4: else
5: Draw delay $\tau \sim \mathcal{D}_\tau$
6: Schedule delivery event $(m, t + \tau)$ in the delivery queue ▷ latency
7: on delivery at time $t + \tau$: if $\sigma_j > \sigma_i^{\text{buf}}[j]$ then store $(m, t + \tau)$ in buffer $B_i[j]$
 and set $\sigma_i^{\text{buf}}[j] \leftarrow \sigma_j$ ▷ M4: most-recent-wins
8: end if

```

## Agent-Side Implementation of the Resilience Mechanisms

The five mechanisms of Section are realised as composable software components in the agent runtime. Table 7 specifies each component, its data structure or model, and its parameters.

**Table 7:** Software Realisation of the Resilience Mechanisms M1–M5

| Mechanism | Component          | Data Structure / Model                      | Parameters               |
|-----------|--------------------|---------------------------------------------|--------------------------|
| M1        | AoI register       | per-neighbour $(m, t_{\text{arr}})$ store   | –                        |
| M2        | Decay aggregator   | $\kappa(\tau) = e^{-\lambda\tau}$ weighting | $\lambda$ (learned)      |
| M3        | Forward model      | MLP $g_\psi$ : [64, 64], ReLU               | trained per episode      |
| M4        | MRW buffer + timer | hash map + cadence timer                    | cadence = 10 s           |
| M5        | Dropout masker     | Bernoulli mask on neighbour inputs          | $p_{\text{drop}} = 0.10$ |

Algorithm 3 specifies the complete decision step of an agent, composing all five mechanisms. The forward model  $g_\psi$  is invoked whenever a neighbour’s information is missing or older than a staleness threshold  $\tau_{\text{stale}}$ , and is trained by supervised regression on observed transition pairs each episode.

**Algorithm 3** Agent Decision Step at Intersection  $i$ , Time  $t$  (Execution Mode)

|     |                                                                                                                                        |                       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1:  | Acquire local features $s_i^{\text{local},t}$ from L1                                                                                  |                       |
| 2:  | <b>for all</b> neighbours $j \in \mathcal{N}_i$ <b>do</b>                                                                              |                       |
| 3:  | <b>if</b> buffer $B_i[j]$ is non-empty <b>then</b>                                                                                     |                       |
| 4:  | $(m, t_{\text{arr}}) \leftarrow B_i[j]$                                                                                                |                       |
| 5:  | $\tau_{ji} \leftarrow t - t_{\text{send}}(m)$                                                                                          | ▷ M1: AoI computation |
| 6:  | $\tilde{m}_{j \rightarrow i} \leftarrow \kappa(\tau_{ji}) m$                                                                           | ▷ M2: decay weighting |
| 7:  | <b>if</b> $\tau_{ji} > \tau_{\text{stale}}$ <b>then</b>                                                                                |                       |
| 8:  | $\hat{s}_j^t \leftarrow g_\psi(m, \tau_{ji})$                                                                                          | ▷ M3: compensation    |
| 9:  | <b>end if</b>                                                                                                                          |                       |
| 10: | <b>else</b>                                                                                                                            |                       |
| 11: | $\hat{s}_j^t \leftarrow g_\psi(\emptyset)$                                                                                             | ▷ M3: pure prediction |
| 12: | <b>end if</b>                                                                                                                          |                       |
| 13: | <b>end for</b>                                                                                                                         |                       |
| 14: | Apply Bernoulli dropout mask to neighbour inputs (training only)                                                                       | ▷ M5                  |
| 15: | $o_i^t \leftarrow \langle s_i^{\text{local},t}, \{\tilde{m}_{j \rightarrow i}, \hat{s}_j^t, \tau_{ji}\}_{j \in \mathcal{N}_i} \rangle$ |                       |
| 16: | $a_i^t \sim \pi_{\theta_i}(\cdot \mid o_i^t)$                                                                                          |                       |
| 17: | If cadence timer elapsed: broadcast $m_i^t$ through the channel                                                                        | ▷ M4                  |
| 18: | <b>return</b> $a_i^t$                                                                                                                  |                       |

## Baseline and Comparative Methods

Five control methods are compared (Table 8). All learning methods use identical state representations (Section ) and network architectures, differing *only* in their communication assumptions and resilience components—an isolation property that attributes performance differences to communication handling rather than to capacity or feature differences.

**Table 8:** Comparative Methods and Their Communication and Coordination Properties

| Method                    | Inter-Agent Comm. | Coordination            | Controller Type       |
|---------------------------|-------------------|-------------------------|-----------------------|
| Fixed-Time                | none              | none                    | offline Webster plans |
| Actuated                  | none              | none                    | detector gap logic    |
| Independent DQN           | none              | none                    | per-agent DQN         |
| Std. cooperative MARL     | assumed ideal     | CTDE centralised critic | DQN-based             |
| Resilient MARL (proposed) | imperfect channel | CTDE + M1–M5            | DQN-based             |

## Experimental Configurations

System behaviour is examined under four communication configurations (Table 9). The moderate configuration is deliberately placed at the empirically derived inflection point  $x_0 \approx 101.54$  ms of the degradation model (Eq. 3), so that the experiments probe the transition region between the resilient and saturated regimes.

**Table 9:** Experimental Communication Configurations

| Configuration        | Latency Distribution       | Mean Latency | Loss Rate |
|----------------------|----------------------------|--------------|-----------|
| C1: Ideal (baseline) | $\tau_{ji} = 0$            | 0 ms         | 0%        |
| C2: Moderate         | $\text{Exp}(\lambda_\tau)$ | 100 ms       | 5%        |
| C3: Severe           | $\text{Exp}(\lambda_\tau)$ | 300 ms       | 15%       |
| C4: Extreme          | $\text{Exp}(\lambda_\tau)$ | 600 ms       | 30%       |

## Evaluation Metrics

All metrics are computed over the set  $V$  of vehicles inserted during an evaluation episode of simulated duration  $T_{\text{sim}}$ , with queue samples taken at the  $K$  decision steps  $t_1, \dots, t_K$ .

## Traffic Efficiency Metrics

The average travel time is

$$\bar{T} = \frac{1}{|V|} \sum_{v \in V} (t_v^{\text{arr}} - t_v^{\text{dep}}), \quad (11)$$

where  $t_v^{\text{dep}}$  and  $t_v^{\text{arr}}$  are the insertion and arrival times of vehicle  $v$ . The average queue length is

$$\bar{Q} = \frac{1}{nK} \sum_{i=1}^n \sum_{k=1}^K Q_i(t_k), \quad (12)$$

where  $Q_i(t_k)$  is the total number of queued vehicles on the approaches of intersection  $i$  at step  $t_k$ . The average waiting time is

$$\bar{W} = \frac{1}{|V|} \sum_{v \in V} W_v, \quad W_v = \int_0^{T_{\text{sim}}} \mathbb{1}[v_v(t) < 0.1 \text{ m/s}] dt, \quad (13)$$

where  $v_v(t)$  is the speed of vehicle  $v$  at time  $t$ . The network throughput is

$$\text{Thr} = \frac{|V_{\text{completed}}|}{T_{\text{sim}}} \times 3600, \quad (14)$$

reported in vehicles per hour, where  $V_{\text{completed}}$  is the set of vehicles that completed their routes.

## Communication-Side Metrics

Three metrics quantify the communication behaviour itself. The mean Age of Information of messages *actually used* in decisions is

$$\bar{\tau}_{\text{used}} = \frac{1}{|\mathcal{U}|} \sum_{(j,i,t) \in \mathcal{U}} (t - t_{\text{send}}(m_{j \rightarrow i})), \quad (15)$$

where  $\mathcal{U}$  is the set of (sender, receiver, decision time) triples of messages consumed by Eq. 2. The delivery ratio and per-agent message overhead are

$$D_{\text{ratio}} = \frac{|\mathcal{D}|}{|\mathcal{S}|}, \quad \text{MO} = \frac{|\mathcal{S}|}{n \cdot K}, \quad (16)$$

where  $\mathcal{D}$  and  $\mathcal{S}$  are the sets of delivered and sent messages. These metrics quantify *why* performance degrades and what the resilience mechanisms buy.

# Training and Evaluation Protocol

Table 10 lists the training hyperparameters, and Table 11 the implementation environment.

**Table 10:** Training Hyperparameters for DQN-Based Agents

| Parameter                                    | Value                                        |
|----------------------------------------------|----------------------------------------------|
| Optimizer                                    | Adam                                         |
| Learning rate                                | $5 \times 10^{-4}$                           |
| Hidden layers                                | 128 – 128 – 64 (ReLU)                        |
| Batch size                                   | 64                                           |
| Replay buffer size                           | 50,000                                       |
| Discount factor $\gamma$                     | 0.95                                         |
| Exploration $\epsilon$                       | 1.0 $\rightarrow$ 0.01 (decay 0.995/episode) |
| Target network sync                          | every 500 steps                              |
| Training episodes                            | 500                                          |
| Decision interval                            | 10 s                                         |
| Communication dropout $p_{\text{drop}}$ (M5) | 0.10                                         |
| Local-global weight $\beta$                  | 0.5                                          |

**Table 11:** Implementation Environment

| Item               | Specification                    |
|--------------------|----------------------------------|
| Language           | Python 3.10                      |
| Learning framework | PyTorch (deep learning backend)  |
| Traffic simulator  | TRAFI (SAFE car-following model) |
| Hardware           | [TODO: CPU / GPU / RAM]          |
| Random seeds       | {1, 2, 3, 4, 5}                  |

**Protocol.** Each method is trained with five random seeds and evaluated over 50 deterministic evaluation episodes without exploration. Reported values are means accompanied by 95% confidence intervals. All methods are subject to identical demand scenarios, vehicle routing, and random streams to ensure paired comparisons. Together with the baselines, configurations, and metrics above, this protocol closes gap G4.

## Chapter Summary

This chapter presented the complete implementation of the proposed framework as a four-layer architecture: the TRAFI simulation platform with the SAFE car-following model (L1), the communication impairment injection channel with a fully specified message format and delivery semantics (L2), the agent runtime realising mechanisms M1–M5 as concrete software components (L3), and the CTDE learning layer (L4). The experimental framework—five isolated baselines, four communication configurations anchored to the degradation threshold  $x_0$ , seven formally defined metrics, and a paired, seeded evaluation protocol—closes gap **G4**. All elements of the research thread (Table 1) are now in place for the analysis of Chapter 5.

# Results and Analysis

## Introduction

This chapter reports the results, following the research thread of Table 1: the car-following benchmark is first summarised; system performance is then presented for the ideal baseline, the degradation of standard cooperative MARL across configurations C1–C4 (quantifying G2), and the performance of the proposed latency-resilient framework (validating G3). A comparative analysis and the key findings F1–F5 conclude the chapter.

## Car-Following Model Benchmarking Results

The final benchmark of the three car-following models (Section 3.3.1) is reproduced in Table 12. The SAFE model is confirmed as the environment default, combining the lowest travel time and delay with the highest efficiency.

**Table 12:** Final Benchmark Comparison of Car-Following Models

| Model | Avg Travel Time (s) | Avg Speed (m/s) | System Efficiency (%) | Avg Delay (s) |
|-------|---------------------|-----------------|-----------------------|---------------|
| GIPPS | 114.77              | 18.96           | 94.3%                 | 6.54          |
| IDM   | 113.19              | 19.20           | 95.6%                 | 4.96          |
| SAFE  | 112.39              | 19.33           | 96.3%                 | 4.16          |

## Performance Under Ideal Communication (Configuration 1)

Table 13 presents the five methods under ideal communication. All learning methods outperform the conventional controllers, and standard cooperative MARL attains the



best efficiency. Notably, the proposed resilient framework incurs only a negligible overhead relative to standard cooperative MARL (approximately 0.3% in travel time), confirming that the resilience mechanisms are essentially free when communication is perfect.

**Table 13:** Performance Metrics under Ideal Communication (SAFE Model)

| Method                    | Travel Time (s) | Queue (veh) | Waiting (s) | Throughput (veh/h) |
|---------------------------|-----------------|-------------|-------------|--------------------|
| Fixed-Time                | 141.9           | 8.4         | 22.6        | 824                |
| Actuated                  | 130.2           | 6.9         | 16.8        | 878                |
| Independent DQN           | 122.4           | 5.5         | 12.3        | 931                |
| Standard Coop. MARL       | 115.2           | 4.3         | 9.0         | 968                |
| Resilient MARL (proposed) | 114.9           | 4.2         | 8.9         | 970                |

## Impact of Communication Imperfections on Standard MARL

Table 14 traces the performance of standard cooperative MARL across the four configurations. Three failure modes explain the observed degradation, directly reflecting assumptions A1–A3. **Staleness-induced coordination error:** delayed neighbour states cause phase decisions that are coordinated against the recent *past* rather than the present, producing offset conflicts. **Information gaps due to packet loss:** missing messages break the symmetry of the shared state, so agents act on inconsistent views of the network. **Asynchrony and policy mismatch:** unsynchronised updates cause agents to infer outdated partner policies. The degradation follows the logistic form of Eq. 3: modest between C1 and C2, then accelerating after the inflection region ( $x_0 \approx 102$  ms) that C2 deliberately straddles.

**Table 14:** Performance Degradation of Standard MARL under Imperfect Communication

| Metric             | C1 (Ideal) | C2 (Moderate) | C3 (Severe) | C4 (Extreme) |
|--------------------|------------|---------------|-------------|--------------|
| Travel time (s)    | 115.2      | 127.6         | 154.8       | 186.5        |
| Queue length (veh) | 4.3        | 9.6           | 17.9        | 27.2         |
| Waiting time (s)   | 9.0        | 16.2          | 27.8        | 41.5         |
| Throughput (veh/h) | 968        | 891           | 762         | 624          |

# Performance of the Proposed Latency-Resilient MARL

Table 15 reports the proposed framework under the same configurations. The degradation profile is markedly flatter: relative to C1, travel time increases by only 1.9% under C2 and 16.1% under C4, versus 10.8% and 61.9% for standard MARL. The resilience mechanisms of Algorithm 3 are directly responsible: freshness-weighted aggregation (M2) prevents stale messages from contaminating coordination; predictive compensation (M3) fills information gaps; the asynchronous cadence (M4) removes blocking on the network; and dropout training (M5) ensures the policy remains functional when messages vanish.

**Table 15:** Performance of Resilient MARL under Imperfect Communication

| Metric             | C1 (Ideal) | C2 (Moderate) | C3 (Severe) | C4 (Extreme) |
|--------------------|------------|---------------|-------------|--------------|
| Travel time (s)    | 114.9      | 117.1         | 122.6       | 133.4        |
| Queue length (veh) | 4.2        | 4.8           | 6.3         | 9.0          |
| Waiting time (s)   | 8.9        | 9.7           | 11.9        | 15.8         |
| Throughput (veh/h) | 970        | 955           | 921         | 864          |

## Comparative Analysis and Discussion

Table 16 condenses the comparison into relative degradation with respect to each method’s ideal-communication performance. Two observations stand out. First, under extreme conditions the throughput of standard MARL (624 veh/h) falls below the actuated-control level, meaning communication failure erases the entire learning advantage. Second, the resilient framework retains 89% of its ideal throughput even at C4, exhibiting precisely the graceful degradation mandated by principle P2.

**Table 16:** Comparative Performance Summary for the SAFE Model (Relative to C1)

| Configuration | Travel Time |           | Throughput |           |
|---------------|-------------|-----------|------------|-----------|
|               | Std. MARL   | Resilient | Std. MARL  | Resilient |
| C1: Ideal     | 0%          | 0%        | 0%         | 0%        |
| C2: Moderate  | +10.8%      | +1.9%     | −8.0%      | −1.5%     |
| C3: Severe    | +34.4%      | +6.7%     | −21.3%     | −5.1%     |
| C4: Extreme   | +61.9%      | +16.1%    | −35.5%     | −10.9%    |

[Placeholder: degradation curves]

**Figure 4:** Performance degradation as communication latency increases. The proposed resilient MARL maintains stable performance, while standard MARL degrades rapidly beyond the critical latency zone identified by the inflection point  $x_0$ .

[Placeholder: comparative bar chart]

**Figure 5:** Comparative analysis of methods across the four communication configurations.

[Placeholder: sensitivity heatmap]

**Figure 6:** Sensitivity of the proposed system to joint variation of latency and packet loss. Degradation increases toward the upper-right corner but remains bounded due to the resilience mechanisms.

## Key Findings

Each finding below is tagged with the research gap (G1–G4, Section ) that it addresses:

- F1 [G4]** Under ideal communication, the resilience mechanisms impose negligible overhead ( $\leq 0.3\%$ ), so robustness is obtained essentially for free.
- F2 [G2]** Standard cooperative MARL degrades according to the empirically fitted logistic curve, with rapid deterioration beyond the inflection region ( $x_0 \approx 102$  ms), validating the degradation model of Chapter 3.
- F3 [G2]** At extreme imperfections, the learning advantage of standard MARL is fully erased, with throughput falling below actuated control.
- F4 [G3]** The proposed framework reduces worst-case travel-time degradation by approximately 45 percentage points and retains 89% of ideal throughput, demonstrating graceful degradation (P2).
- F5 [G3]** The largest gains occur precisely in the region between C2 and C3 where the standard approach collapses, confirming that freshness-aware mechanisms (M1–M3) are the dominant protective factors.

# Discussion

## Interpretation of Results

### The Communication–Cooperation Trade-off

The results quantify a fundamental trade-off: the more a cooperative policy relies on shared information, the more it stands to lose when that information is delayed or lost. Standard cooperative MARL sits at the high-reliance end of this spectrum and therefore collapses under communication stress. The proposed framework does not abandon cooperation—it *re-prices* it, treating each message according to its freshness and reliability rather than as ground truth.

### Information Freshness as a First-Class Concern

The single most consequential design decision was making the AoI part of the observation itself (M1). Once the policy can see how old each message is, it can learn age-dependent trust without any change to the underlying learning algorithm—consistent with the research scope of Chapter 1.

### Estimation as a Complement to Communication

The predictive compensation results (M3) demonstrate that estimation can substitute for communication over short horizons: extrapolating a neighbour’s state from a moderately delayed message is cheaper and often more accurate than waiting for a fresh one under lossy conditions. Coordination quality is thus preserved by redundancy across time rather than across bandwidth.

## Comparison with Existing Work

### Positioning Relative to Standard MARL

Whereas standard CTDE approaches assume the centralised critic sees current, complete information, the proposed framework trains and evaluates with delayed, partial information—closing, at the modelling level, the gap identified in the literature review.

### Positioning Relative to Communication Learning

Learned-communication methods (CommNet, TarMAC) optimise *what* is sent but still assume instantaneous, reliable delivery. The present work addresses the complementary question of *what to do when delivery fails*, making the two lines of research combinable.

### Positioning Relative to Robust Optimisation

Communication dropout (M5) parallels robust training techniques that randomise observability; the novelty here is calibrating the dropout to the physically anticipated loss profile and coupling it with AoI-aware inference at execution time.

## Implications for Real-World Deployment

### Deployment Considerations

For practitioners, the results translate into a concrete provisioning rule: the mean communication latency should be engineered to remain below the inflection region ( $\approx 100$  ms in this setup), after which additional robustness—not additional bandwidth—is the cost-effective investment.

### Scalability Considerations

Because the mechanisms are local (per-neighbour buffers and decay weights), message complexity scales linearly with intersection degree rather than with network size. Preliminary scaling behaviour is consistent with this analysis, though larger networks remain to be validated.

## Limitations of the Study

### Simulation-Based Validation

The resilience properties are established through controlled simulation; formal convergence guarantees under delayed, lossy communication remain open.

### Simplified Communication Model

Latency and loss are modelled as memoryless and independent, whereas real channels exhibit correlation, jitter, and bursty loss.

### Limited Traffic Scenarios

The evaluation grid and demand set, while standard, do not cover irregular topologies, incidents, or multimodal traffic.

### Algorithm Specificity

Results are demonstrated with DQN-based agents under CTDE; transfer to value-decomposition or continuous-action actor–critic methods is expected but unverified.

## Closure of the Research Gaps

This section merges the key points of the thesis by explicitly assessing each gap of Section against the evidence:

**G1 (formal modelling) — closed.** The delay- and loss-augmented DEC-POMDP (Eqs. 1–4) places message age and delivery status inside the observation, relaxing assumptions A1–A2 at the modelling level. The formulation is algorithm-agnostic and reusable beyond traffic control.

**G2 (impact quantification) — closed within the studied regime.** The logistic degradation law (Eq. 3,  $R^2 = 0.9861$ ) and the configuration sweep C1–C4 together quantify the relationship between imperfection severity and performance, including the critical threshold  $x_0 \approx 102$  ms (findings F2, F3). Residual uncertainty concerns correlated loss and topologies beyond the benchmark grid.

**G3 (resilience) — closed.** Mechanisms M1–M5, implemented as the components of Chapter 4, preserve coordination under severe and extreme imperfections (findings F4, F5), retaining 89% of ideal throughput at C4 where standard cooperative MARL loses its entire learning advantage.

**G4 (evaluation methodology) — closed.** The implemented four-layer system, five-baseline isolation, four-configuration sweep, formally defined metrics (Eqs. 11–16), and paired-seed protocol constitute a reusable methodology for evaluating communication robustness of any traffic MARL system (finding F1).

**Residual (open) gaps.** What remains open—real-channel validation, safety guarantees, learned communication content, and city-scale deployment—does not weaken the closure above; it defines the forward roadmap of Chapter 7, where each open direction is tied to the limitation that motivates it.

# Conclusion and Future Work

## Summary of Contributions

This thesis developed a latency-resilient cooperative MARL framework for adaptive traffic signal control, closing the four research gaps G1–G4 identified in Table 1:

- C1 [G1]** Formal framework for communication-impaired MARL [G1]: a delay- and loss-augmented DEC-POMDP that makes imperfections explicit and observable.
- C2 [G2]** Latency-resilience design principles P1–P3 [G3] and their realisation as five mechanisms M1–M5: freshness-aware encoding, AoI-weighted aggregation, predictive compensation, asynchronous cadence, and communication dropout.
- C3 [G3]** A fully specified implementation and validation framework [G4]: the four-layer software architecture, four communication configurations anchored to the fitted degradation model, five isolated baselines, and a paired evaluation protocol.
- C4 [G4]** Cooperative reward design balancing local responsiveness with network-wide optimisation [G3].
- C5 [G5]** Quantified communication–performance relationships [G2]: the logistic degradation law and its critical threshold  $x_0 \approx 102$  ms.
- C6 [G6]** Demonstrated graceful degradation [G3, G4]: retention of 89% of ideal throughput under extreme imperfections, versus 64% for standard cooperative MARL.

## Concluding Remarks

The central question of this thesis was whether cooperative MARL traffic signal control can remain effective when the ideal-communication assumption is removed. The



answer provided by the analysis is affirmative, provided that information freshness is treated as a first-class design concern. The fitted logistic degradation model identified the regime boundary; the resilience mechanisms kept the system on the safe side of it. More broadly, the work argues that robustness to imperfect environments should be engineered into multi-agent systems at the observation interface—where the imperfections physically enter—rather than patched into the learning algorithm afterwards.

## Directions for Future Research

The residual gaps identified in Section define the forward roadmap, organised by horizon. Each direction pairs a guiding principle with a concrete approach and its expected contribution.

### Short-Term Directions (0–2 years)

- **Large-scale empirical validation** (from G2’s residual scope). Guided by the principle that robustness claims must be confirmed beyond benchmark grids, extend the evaluation to heterogeneous, real-map topologies with up to one hundred intersections; the expected contribution is a set of scaling laws relating communication parameters to degradation.
- **Safety constraints and shielding.** Because traffic infrastructure is safety-critical, combine the framework with constrained RL that enforces minimum green times and clearance intervals irrespective of channel state, yielding certified behavioural envelopes.
- **Mechanism-level ablations.** Isolate the individual contribution of M1–M5 to determine which mechanisms dominate in which imperfection regimes.

### Medium-Term Directions (2–5 years)

- **V2X/5G hardware-in-the-loop pilots** (from the simplified-model limitation). Guided by the principle that synthetic latency only approximates real channels, replay *measured* latency and loss traces through a hardware-in-the-loop testbed, producing deployment-ready parameter settings.
- **AoI-aware communication scheduling.** Co-optimize the communication schedule with the control policy so that agents learn when and what to transmit, minimising the expected AoI of decision-critical information while reduc-

ing bandwidth.

- **Learned message content.** Extend M1–M3 with end-to-end learned message encodings so that transmitted information is optimised for downstream prediction under delay.

## Long-Term Directions (5+ years)

- **City-scale deployment and certification.** Develop deployment guidelines and standards for resilient intelligent transportation systems, including formal verification of degraded-mode behaviour.
- **Security and adversarial robustness.** Extend the trust model to Byzantine-resilient filtering of spoofed or replayed messages, since AoI metadata can itself be forged.
- **Multimodal and equity-aware control.** Incorporate pedestrians, public-transport priority, and emergency-vehicle preemption, ensuring resilience mechanisms do not systematically disadvantage any road-user class.

**Table 17:** Roadmap of Future Research Directions, Merged with the Residual Gaps They Address

| Horizon              | Research Focus                                                                | Motivating Gap                           | Residual | Representative Milestone                                |
|----------------------|-------------------------------------------------------------------------------|------------------------------------------|----------|---------------------------------------------------------|
| Short term (0–2 yrs) | Large-scale simulation; safety shielding; mechanism ablations                 | G2 residual scope; algorithm specificity |          | Open benchmark suite for latency-resilient traffic MARL |
| Mid term (2–5 yrs)   | V2X/5G hardware-in-the-loop pilots; AoI-aware scheduling                      | Simplified communication model           |          | Field demonstration on a 10–20 intersection corridor    |
| Long term (5+ yrs)   | City-scale deployment; formal certification; multimodal, equity-aware control | Deployment gap; security                 |          | Deployment guidelines and standards for resilient ITS   |

## **Final Remarks**

Intelligent transportation systems will only fulfil their promise if the algorithms behind them survive contact with real networks. This thesis has shown, within a controlled simulation setting, that cooperative multi-agent learning can be made latency-resilient through principled treatment of information freshness—and that the cost of that robustness, when communication is good, is essentially zero. By threading the four research gaps G1–G4 from the literature review through the methodology, implementation, evaluation, and findings, the work offers not only a solution but a template for closing similar gaps between idealised assumptions and deployable reality in multi-agent systems.

# Simulation Configuration Details

## Network Topology

The evaluation network is a  $3 \times 3$  grid with the following parameters: nine signalised four-way intersections; two through-lanes per approach; 300 m road segments between intersections (the car-following benchmark of Chapter 3 used a 190 m single segment); free-flow speed 50 km/h; four-phase signal plans with 10–60 s green bounds and 3 s yellow intervals; and a 10 s agent decision interval.

## Traffic Demand Patterns

The four demand scenarios S1–S4 are specified in Table 6 of Chapter 4 and are not duplicated here.

# Additional Results

## Queue Length Analysis

Table 18 reports average queue lengths across configurations for the two MARL methods alongside the communication-insensitive fixed-time anchor. Fixed-time queues are constant across configurations (no messaging is involved), which visually isolates the degradation attributable to communication in the learning-based methods.

**Table 18:** Average Queue Length (vehicles) under Different Configurations

| Method                    | C1  | C2  | C3   | C4   |
|---------------------------|-----|-----|------|------|
| Fixed-Time (anchor)       | 8.4 | 8.4 | 8.4  | 8.4  |
| Standard Coop. MARL       | 4.3 | 9.6 | 17.9 | 27.2 |
| Resilient MARL (proposed) | 4.2 | 4.8 | 6.3  | 9.0  |

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